

Operating Systems 2025 Spring Final Exam (2025/06/04)

Please answer questions in order and leave blank for unanswered questions

There are 18 questions

1. (8%) What's important transparencies of distributed file systems? Please list at least four and briefly explain each. (2% each)

Ans.

(a) Access transparency: client programs should not be aware of the distribution of files. The same APIs are provided for accessing local and remote files.

(b) Location transparency: client programs should see an uniform file name space. Files may be relocated without changing their names. Programs see the same name space wherever they are executed.

(c) Mobility transparency: both client programs and administration configurations in client nodes are unchanged on file relocations.

(d) Performance transparency: the performance of client programs should not be affected when the workload varies within a small range.

(e) Scaling transparency: the DFS can be semi-linearly adjusted to deal with a wide range of loads and network sizes.

2. (4%) What are the main purposes of a file? Please list at least two and briefly explain each. (2% each)

Ans.

(a) Permanent storage of information. Files are stored on disks or other non-volatile storage media. Files contain both data and metadata. The data consists of a sequence of items that are accessible by operations such as read and write. The metadata contains information such as the length of the file, timestamps, file type, owner's identity, access control lists, and more.

(b) Sharing of information. A file can be created by one application and then shared with different applications at a later time.

3. (6%) How does the deadlock for Maekawa's mutual exclusion Algorithm happen? Please give an example to demonstrate it and provide a solution to avoid this.

Ans.

Deadlock example (2%):

Consider three processes, p_1 , p_2 , and p_3 , where $V_1=\{p_1, p_2\}$, $V_2=\{p_2, p_3\}$, and $V_3=\{p_3, p_1\}$. If they concurrently request to enter the CS, then it is possible for p_1 to reply to itself and hold off p_2 , for p_2 to reply to itself and hold off p_3 , and for p_3 to reply to itself and hold off p_1 . Each process has received only one replies and none can proceed.

Solution to avoid deadlock (4%):

Deadlock handling requires three extra types of messages:

A FAILED message from p_i to p_j indicates that p_i cannot grant p_j a vote because it has voted to a process with a higher priority request.

An INQUIRE message from p_i to p_j indicates that p_i would like to find out from p_j if it has all votes from processes in V_j .

A YIELD message from p_i to p_j indicates that p_i is returning the vote to p_j for yielding to a higher priority request at p_j .

To avoid deadlock, a process p needs to yield a vote if the timestamp of p 's request is larger than the timestamp of some other requests waiting for the same vote.

The inquire and yield process need to be added. When the process has granted some other lower priority process and the incoming higher priority process comes, the inquiry is sent to the granted process. The granted process checks the existence of received failed and the non-replied yield. It sends yield back to the deciding process and let the process put it into the queue and proceed the incoming process.

4. (6%) Please give three possible cache location policies and their advantages. (2% each)

Ans.

(a) Server memory: easy implementation and transparent to clients.

(b) Client memory: provide good performance speed up.

(c) Client disk: can cache more and larger files.

5. (6%) Please briefly describe the concept of "happened before" relation as defined by Lamport. (2% each)

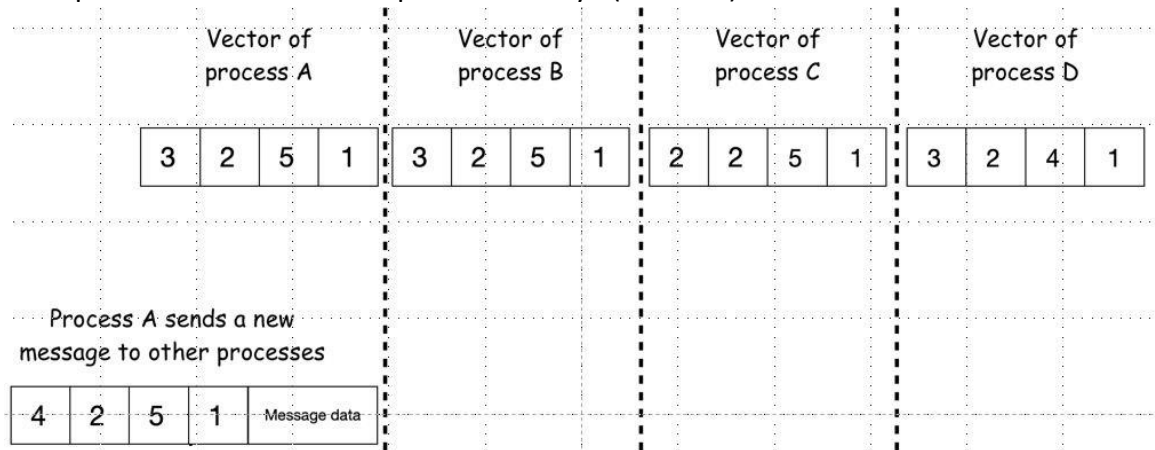
Ans: The "happened before" relation ($\rightarrow$) is defined by three rules:

HB1 states that if two events occur in the same process, the one that occurs earlier happened before the later one.

HB2 states that the sending of a message happened before the receiving of that message.

HB3 states that if event 'e' happened before 'e'' and 'e'' happened before 'e''' , then 'e' happened before 'e''' (transitivity).

6. (6%) For the figure below, which processes can receive the new message sent from process A in the CBCAST protocol? Why? (2% each)



Ans.

In CBCAST, If $S[i]=R[i]+1$ and for all $j \neq i$, $S[j] \leq R[j]$, then the message is delivered to the application, otherwise, the message is kept in the buffer.

- (a) Process B can receive the message sent from process A because $A[1]=B[1]+1$ and $A[2]=B[2]$, $A[3]=B[3]$, $A[4]=B[4]$.
- (b) Process C can't receive the message because the condition $A[1] \neq C[1]+1$.
- (d) Process D can't receive the message because the condition $A[3] > D[3]$.

7. (4%) Please briefly describe the primary roles of the NameNode in HDFS? (2% each)

Ans.

The NameNode in HDFS is responsible for file system **namespace (metadata) management**, **name-to-file mapping**, **handling operations like open, close, and renaming**, and **data blocks allocation/deallocation**.

8. (6%) Please give 3 differences between replication and caching. (2% each)

Ans.

- (a) A replica is a copy in a server disk and a cache is a copy of a replica.
- (b) A cache is dependent on the locality in file access patterns.
- (c) A replica depends on availability and performance requirements.
- (d) A replica is usually persistent and widely known.
- (e) The existence of a cache depends on the presence of a replica.

9. (6%) Please list the three synchronization modes of NTP servers and briefly explain each. (2% each)

Ans.

- (a) Multicast mode is intended for use on a high-speed LAN.

One or more servers periodically multicasts their clock values to other hosts connected by the LAN. Upon receiving a clock value, a host sets its clock by assuming a small delay. This mode can achieve only relatively low accuracies, but it is considered sufficient for many purposes.

(b) Procedure-call mode is similar to the Cristian's algorithm. In this mode, one server accepts requests from other computers. Upon receiving a request, it replies its current clock reading. It can achieve higher accuracy that multicast can not support.

(c) Symmetric mode is intended for servers in the lower strata. It can achieve the highest accuracy. A pair of servers operating in symmetric mode exchange messages bearing timing information.

10. (4%) What are the design characteristics of AFS for achieving high scalability? Please briefly describe these characteristics?

Ans. (2% each)

AFS has two unusual design characteristics to achieving high scalability.

1. Whole file serving (4%): The entire contents of directories and files are transmitted to client computers by AFS servers. In AFS 3, files larger than 64 Kbytes are transferred in 64 Kbyte chunks.
2. Whole file caching (6%): Once a copy of a file has been transferred to a client computer it is stored in a cache on the local disk. The cache contains several hundred of the files most recently used on the client computer. The cache is permanent and can survive reboots of the client computer. Local copies of files are used to satisfy clients' open requests in preference to remote copies whenever possible.

11. (6%) Please give and describe at least 3 file sharing semantics of distributed file systems. (2% each)

Ans.

(a) One copy update semantics (UNIX): the result of every write operation must be visible immediately.

(b) Session update semantics (AFS): only after closing the updated file, the result of write operations becomes available.

(c) Append only update semantics (HDFS): data is never modified once it has been written to a file, and new data can only be appended to the existing file.

(d) Copy on write update semantics: In a copy-on-write file system, when a process requests to modify a shared file, a copy of the original file is created for the requester if another process has already modified or holds a reference to it. Otherwise, the original file is considered immutable, and the requesting process can safely modify the copy without affecting other processes.

12. (6%) Discuss whether the following operations are idempotent: (2% each)
- (a) Pressing an elevator call button
 - (b) Sorting a list
 - (c) Appending to a file
- Ans.
- (a) Idempotent (elevator stays in requested state)
 - (b) Idempotent (sorting a sorted list does not change the sorted list)
 - (c) Non-idempotent (two appends will add twice as much data)
13. (4%) Assuming there are N processes, describe the best and worst cases in the Bully algorithm and how many messages are required in each case. (2% each)
- Ans.
- (a) Worst Case: The lowest priority process initiates election. It requires $O(N^2)$ messages
 - (b) Best Case: The eventual leader initiates election. It requires $O(N)$ messages
14. (6%) In algorithms for Distributed Mutual Exclusion (DME), what are the essential requirements for mutual exclusion? (2% each)
- Ans.
- ME1: (safety) At most one process may execute in the CS at a time.
- ME2: (liveness) Requests to enter and exit the CS eventually succeed.
- ME3: (HB ordering) Entry to the CS is granted in the HB order of requests.
15. (4%) What are client-initiated and server initiated approaches in cache validation policies? (2% each)
- Ans.
- (a) Client initiated approach. A client initiates a validity check with the server. Three possible checking variations are:
 - Checking before every access. (Unix semantics)
 - Periodic checking.
 - Checking when opens a file. (session semantics)The validation is usually based on timestamp or version#.
 - (b) Server initiated approach. The server records cache information for each client. When the server detects a potential inconsistency, it reacts.
16. (4%) Please briefly describe the stateless server and the stateful server? (2% each)
- Ans.
- (a) A stateless server does not keep any state information of clients.
 - (b) A stateful server maintains clients' state information from one access request to the next.

17. (6%) Please list the three variations of the "Delayed write" scheme. (2% each)

Ans.

(a) Write on ejection. (victim of cache replacement)

(b) Periodic write.

(c) Write on close. (session semantics)

18. (8%) Please give and describe at least 4 multi-copy update policies. (2% each)

Ans.

(a) Read Only Replication (ROR). Only immutable files are replicated to avoid consistency issue.

(b) Read One Write All Protocol (R1Wn). A read OP is performed by reading any copy of the file. A write OP is performed by writing to all replicas of the file.

Some form of locking has to be used to carry out a write operation. It is suitable for implementing UNIX like semantics. It is very efficient when the read/write ratio is large.

(c) Available Copies Protocol. A write OP is performed by writing to all available copies. On recovery from a failure, a server must 1st brings itself up to date by copying from other servers before accepting any user request. It can tolerate site crashes.

(d) Primary Copy Protocol. For each replicated file, one copy is designated as the primary copy and all the others are secondary copies. Read OPs can be performed using any copy. All write OPs are 1st performed only on the primary copy. Secondary copies are then updated later.

(e) Quorum Based Protocols. Assume that a replicated file F has a total of n copies. For each read OP, a set of r replicas (read quorum) must be contacted to get the latest version. For each write OP, a set of w replicas (write quorum) must be locked to perform the updates. If $(r+w > n \text{ and } 2w > n)$ then every read OP always gets the result of the latest write and every write OP is always performed on top of the latest write.